

DEEPA HANDMADE

EXPANDED DETAILED FUNCTIONAL REQUIREMENTS DOCUMENT FRD v3.0 — Module-Level Implementation & UAT Reference

Attribute	Value
Product	Deepa Handmade
Business Model	Retail + Custom Orders + Wholesale / Bulk
Product Scope	Handmade wire / plastic-wire baskets, bags and related handmade products
Payment	Manual UPI verification + Cash on Delivery only
Core Modules	18
Document Depth	Screen + field + action + workflow + rule + exception + permission + acceptance
Primary Users	Customer, Sales/Support, Production, QC, Packing, Admin, Super Admin

This version expands all 18 modules. It is intended as the functional baseline for UI/UX, development, QA and UAT. API route names, exact database schema and infrastructure are finalized separately in technical design.

Document Control

Version	Status	Purpose
v1.0	Draft	Initial detailed functional outline
v2.0	Draft	Expanded implementation-level FRD
v3.0	Expanded Draft	All 18 modules expanded with module-specific functional detail, validations, states, exceptions and UAT criteria

Global Functional Rules

- Server is authoritative for price, stock, payment state, order state and permissions.
- Every important entity has a unique identifier.
- Historical order/payment snapshots must remain stable after catalogue changes.
- Critical state changes record actor and timestamp.
- Uploaded files are validated and securely stored.
- Soft deactivation is preferred where historical records depend on a record.
- Client-side validation improves UX; server-side validation is mandatory.

Module 1 — Authentication & User Management

Identity, account lifecycle, profiles, staff users, roles and permissions.

A. Screens / Components

- Registration
- Login
- Logout
- Forgot password
- Reset password
- Profile
- Address book
- Admin login
- Staff management
- Role manager
- Permission matrix
- Account status

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Full Name	Text	Yes	2-100 chars	Module-specific validation; 2-100 chars
Mobile	String	Yes	Valid; uniqueness rule	Module-specific validation; Valid; uniqueness rule
Email	String	No	Valid email	Module-specific validation; Valid email
Password	Secret	Yes	Configured password policy	Module-specific validation; Configured password policy
Role	Enum	Staff	Configured role	Module-specific validation; Configured role
Status	Enum	Yes	Active/Suspended/Disabled	Module-specific validation; Active/Suspended/Disabled
Profile Image	File	No	Image validation	Module-specific validation; Image validation
Created/Updated	DateTime	System	Audit timestamps	Module-specific validation; Audit timestamps

C. User & System Actions

- Register/login
- Reset password
- Edit profile
- Add/edit address
- Create staff
- Assign role
- Disable user
- Revoke sessions

D. Workflow & State Behavior

Register → optional verification → active → login → session → logout
Admin login → permission resolution → dashboard
Disable/suspend → active sessions revoked according to security policy

E. Business Rules

- Duplicate mobile/email
- Suspended account cannot login
- Customer cannot change own role
- Reset token single-use
- At least one Super Admin must remain

F. Exceptions / Error Handling

- Rate-limit authentication
- Hash passwords
- Server-side authorization
- Audit role/status changes

G. Security / Permissions

- Valid registration works
- Wrong password rejected
- Suspended user blocked
- Role restrictions enforced
- Reset works

- Admin changes audited

H. Acceptance Criteria

- Primary Authentication & User Management workflow completes successfully with valid data.
- Mandatory fields and boundary values are validated.
- Unauthorized access/action is blocked.
- Duplicate submission/retry behavior is safe.
- Persisted data matches the approved business action.
- Relevant status/history/audit information is available.

I. QA Test Focus

- Positive path, negative path and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Mobile/responsive behavior for customer-facing screens where applicable.
- Audit/log verification for sensitive actions.

Module 2 — Customer Website & CMS

Public storefront, content pages, banners, navigation and business information.

A. Screens / Components

- Home
- Header/footer
- Hero banner
- Category navigation
- Product listing
- About
- Contact
- FAQ
- Custom Orders
- Wholesale
- Policy pages
- CMS editor
- Banner manager

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Page ID	System	Yes	Unique	Module-specific validation; Unique
Page Type	Enum	Yes	Configured	Module-specific validation; Configured
Title	Text	Yes	Required	Module-specific validation; Required
Slug	Text	Yes	Unique URL-safe	Module-specific validation; Unique URL-safe
Content	Rich/Text	Yes	Sanitized	Module-specific validation; Sanitized
Banner	File	Conditional	Validated	Module-specific validation; Validated
CTA URL	URL	No	Safe destination	Module-specific validation; Safe destination
SEO Title	Text	No	Metadata	Module-specific validation; Metadata
SEO Description	Text	No	Metadata	Module-specific validation; Metadata
Status	Enum	Yes	Draft/Published/Archived	Module-specific validation; Draft/Published/Archived

C. User & System Actions

- Create/edit page
- Preview
- Publish
- Unpublish
- Archive
- Reorder banners
- Configure CTA
- Update contact information

D. Workflow & State Behavior

Draft → validation → preview → publish
Published → edit → republish
Archived content remains retained but not public

E. Business Rules

- Drafts not public
- Slug unique
- Invalid media blocked
- Broken CTA prevented

F. Exceptions / Error Handling

- CMS access role protected
- Rich content sanitized
- Private drafts not exposed

G. Security / Permissions

- Published content visible
- Draft hidden
- Banner order correct
- CMS changes reflected without deployment

H. Acceptance Criteria

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Module 3 — Product & Category Management

Catalogue master data, categories, products, SKU, pricing, MOQ, media and availability.

A. Screens / Components

- Category list/form
- Product list/form
- Product form
- Media manager
- Pricing
- Featured products
- Stock visibility
- Activation

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Category Name	Text	Yes	Unique within parent	Module-specific validation; Unique within parent
Product Name	Text	Yes	Required	Module-specific validation; Required
SKU	String	Yes	Globally unique	Module-specific validation; Globally unique
Description	Rich/Text	Yes	Sanitized	Module-specific validation; Sanitized
Base Price	Decimal	Yes	>=0	Module-specific validation; >=0
Sale Price	Decimal	No	Business validation	Module-specific validation; Business validation
MOQ	Integer	Yes	>=1	Module-specific validation; >=1
Weight	Decimal	No	>0	Module-specific validation; >0
Status	Enum	Yes	Draft/Active/OOS/Inactive/Archived	Module-specific validation; Draft/Active/OOS/Inactive/Archived
Primary Image	File/Ref	Yes	Validated	Module-specific validation; Validated

C. User & System Actions

- Create/edit category
- Create/edit product
- Upload/reorder images
- Set price
- Set MOQ
- Feature
- Activate/deactivate

D. Workflow & State Behavior

Draft → complete mandatory data → activate → sellable
Active → OOS according to stock policy → restock → sellable

E. Business Rules

- SKU unique
- Inactive products cannot be newly ordered
- MOQ enforced
- Historical orders keep snapshot
- Price changes auditable

F. Exceptions / Error Handling

- Duplicate SKU
- Invalid price
- Missing category
- Invalid image
- MOQ/stock conflict

G. Security / Permissions

- Catalog writes restricted
- Media upload secured
- Price changes audited

H. Acceptance Criteria

- Activation requires mandatory data
- Correct listing
- Invalid price blocked
- Historical order unchanged

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 4 — Product Variants & Customization

Color, size, handle, pattern and custom requirements.

A. Screens / Components

- Variant master
- Variant matrix
- Customer selector
- Custom size
- Custom color/design
- Handle
- Notes
- Reference image

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Option Type	Enum/Text	Yes	Color/Size/Handle/etc.	Module-specific validation; Color/Size/Handle/etc.
Option Value	Text	Yes	Active value	Module-specific validation; Active value
Variant SKU	String	No	Unique if used	Module-specific validation; Unique if used
Price Delta	Decimal	No	Configured	Module-specific validation; Configured
Variant Stock	Integer	No	If tracked	Module-specific validation; If tracked
Custom Size	Text	No	Unit-aware	Module-specific validation; Unit-aware
Custom Note	Text	No	Length limit	Module-specific validation; Length limit
Reference Image	File	No	Validated	Module-specific validation; Validated

C. User & System Actions

- Add option
- Enable/disable
- Set variant price
- Set variant stock
- Select variant
- Add note
- Upload reference
- Submit custom request

D. Workflow & State Behavior

Product → options → validate combination → calculate price → cart
Non-standard requirement → custom request → quotation

E. Business Rules

- Only enabled options visible
- Combination unique
- Price-impacting custom work uses quotation
- Variant snapshot stored in order

F. Exceptions / Error Handling

- Unavailable combination
- Invalid size
- Unsupported file
- Missing variant price

G. Security / Permissions

- Customer file access controlled
- Admin controls variant master

H. Acceptance Criteria

- Selected options stored
- Price correct
- Custom requirements retained
- Invalid combination blocked

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.

- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 5 — Search, Filter & Wishlist

Product discovery and customer saved products.

A. Screens / Components

- Search
- Results
- Category filter
- Price filter
- Color
- Size
- Availability
- Sort
- Wishlist

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Query	String	No	Trim/sanitize	Module-specific validation; Trim/sanitize
Category	ID	No	Active	Module-specific validation; Active
Min/Max Price	Decimal	No	Valid range	Module-specific validation; Valid range
Color	Value	No	Active option	Module-specific validation; Active option
Size	Value	No	Active option	Module-specific validation; Active option
Availability	Enum	No	In stock/all	Module-specific validation; In stock/all
Sort	Enum	No	Configured	Module-specific validation; Configured
Wishlist	Reference	Yes	Customer-owned	Module-specific validation; Customer-owned

C. User & System Actions

- Search
- Filter
- Clear
- Sort
- Add/remove wishlist
- Open product

D. Workflow & State Behavior

Query/filter → result → product → wishlist/cart

E. Business Rules

- Only active public products
- Wishlist customer-specific
- Inactive item handled by policy

F. Exceptions / Error Handling

- No results
- Invalid filters
- Deactivated wishlist item

G. Security / Permissions

- No private data in search
- Wishlist access owner-only

H. Acceptance Criteria

- Results relevant
- Filters combine
- Wishlist persists

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 6 — Cart Management

Purchase basket, quantity control, price and stock revalidation.

A. Screens / Components

- Cart
- Mini cart
- Quantity
- Remove
- Clear
- Validation messages

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Cart ID	System	Yes	Unique	Module-specific validation; Unique
Owner	Customer/session	Yes	Access check	Module-specific validation; Access check
Product/Variant	Reference	Yes	Valid	Module-specific validation; Valid
Quantity	Integer	Yes	>= MOQ	Module-specific validation; >= MOQ
Unit Price	Decimal	System	Revalidated	Module-specific validation; Revalidated
Discount	Decimal	System	Validated	Module-specific validation; Validated
Subtotal	Decimal	System	Calculated	Module-specific validation; Calculated

C. User & System Actions

- Add
- Merge
- Update quantity
- Remove
- Clear
- Validate
- Checkout

D. Workflow & State Behavior

Add → validate → persist → edit → final validation → checkout

E. Business Rules

- Server is source of truth
- Stock/price revalidated
- MOQ enforced
- Duplicate requests idempotent

F. Exceptions / Error Handling

- Stock changed
- Price changed
- Product inactive
- Coupon invalid

G. Security / Permissions

- Cart ownership enforced
- No client-controlled final price

H. Acceptance Criteria

- Totals correct
- Invalid quantity blocked
- Out-of-stock blocked
- Duplicate add deterministic

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 7 — Checkout & Address

Delivery details, shipping, discount, order summary and payment selection.

A. Screens / Components

- Address list
- Add/edit address
- Shipping
- Discount
- Summary
- Payment selection
- Place order

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Recipient	Text	Yes	Valid	Module-specific validation; Valid
Mobile	String	Yes	Valid	Module-specific validation; Valid
Address	Text	Yes	Complete	Module-specific validation; Complete
City/Town	Text	Yes	Required	Module-specific validation; Required
District	Text	Yes	Required	Module-specific validation; Required
State	Text	Yes	Required	Module-specific validation; Required
Pincode	String	Yes	Indian PIN	Module-specific validation; Indian PIN
Landmark	Text	No	Optional	Module-specific validation; Optional
Shipping Method	Enum	Yes	Configured	Module-specific validation; Configured
Payment Method	Enum	Yes	UPI/COD	Module-specific validation; UPI/COD
Terms	Boolean	Conditional	Business policy	Module-specific validation; Business policy

C. User & System Actions

- Select/add address
- Calculate shipping
- Apply coupon
- Review
- Select payment
- Place order

D. Workflow & State Behavior

Cart validation → address → shipping → discount → final total → payment → order

E. Business Rules

- Address snapshot stored
- Shipping server-calculated
- Only enabled UPI/COD
- Order creation idempotent

F. Exceptions / Error Handling

- Invalid PIN
- Incomplete address
- Shipping unavailable
- Coupon invalid
- Stock changed
- Method disabled

G. Security / Permissions

- Address owner-only
- Final amount server-calculated

H. Acceptance Criteria

- Missing fields block
- Stored order equals summary
- Only valid methods appear

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.

- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 8 — Payment Management

Manual UPI payment verification and COD; no payment gateways in initial scope.

A. Screens / Components

- Payment method
- UPI instructions
- QR
- UPI ID
- Amount
- Proof upload
- UTR
- Verification queue
- Approve
- Reject
- Resubmit
- COD confirmation

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Payment ID	System	Yes	Unique	Module-specific validation; Unique
Order ID	Reference	Yes	Linked	Module-specific validation; Linked
Method	Enum	Yes	UPI/COD	Module-specific validation; UPI/COD
Amount	Decimal	Yes	Order payable	Module-specific validation; Order payable
UPI ID	String	UPI	Settings	Module-specific validation; Settings
UPI QR	File	UPI	Settings	Module-specific validation; Settings
Screenshot	File	UPI	Required	Module-specific validation; Required
UTR	String	UPI	Configurable required	Module-specific validation; Configurable required
Status	Enum	Yes	Pending/Submitted/Review/Approved/Rejected	Module-specific validation; Pending/Submitted/Review/Approved/Rejected
Rejection Reason	Text	Reject	Mandatory	Module-specific validation; Mandatory
Verified By	User	Approve	Authorized	Module-specific validation; Authorized
Verified At	DateTime	Approve	System	Module-specific validation; System

C. User & System Actions

- Show QR/ID
- Upload proof
- Submit proof
- View status
- Approve
- Reject
- Request resubmission
- Confirm COD

D. Workflow & State Behavior

UPI: Order → pending → pay externally → proof → review → approve/reject
Approve → paid + order confirmed
Reject → reason + resubmission
COD → COD confirmed → order confirmed

E. Business Rules

- Screenshot never equals paid
- Only authorized approval confirms
- Reject requires reason
- Approved amount reconciles
- Historical attempts retained
- Disabled UPI affects new checkout only

F. Exceptions / Error Handling

- Wrong amount
- Unreadable proof
- Duplicate UTR
- Missing proof

- UPI disabled
- Unauthorized approval

G. Security / Permissions

- Proof restricted
- Approval audited
- File upload secured

H. Acceptance Criteria

- End-to-end UPI works
- Approve/reject
- Resubmit
- COD no proof
- Verifier/time recorded

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 9 — Order Management

Commercial order record, status lifecycle, cancellation, timeline and operational handoffs.

A. Screens / Components

- Order list
- Detail
- Timeline
- Customer history
- Status actions
- Cancellation
- Notes

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Order ID	System	Yes	Unique	Module-specific validation; Unique
Customer	Reference	Yes	Linked	Module-specific validation; Linked
Items	Snapshot	Yes	Immutable	Module-specific validation; Immutable
Subtotal	Decimal	Yes	Stored	Module-specific validation; Stored
Discount	Decimal	Yes	Stored	Module-specific validation; Stored
Shipping	Decimal	Yes	Stored	Module-specific validation; Stored
Total	Decimal	Yes	Stored	Module-specific validation; Stored
Payment Status	Enum	Yes	Controlled	Module-specific validation; Controlled
Order Status	Enum	Yes	State machine	Module-specific validation; State machine
Address Snapshot	Object	Yes	Immutable	Module-specific validation; Immutable
Cancel Reason	Text	Conditional	Required if cancelled	Module-specific validation; Required if cancelled
Internal Note	Text	No	Staff only	Module-specific validation; Staff only

C. User & System Actions

- View
- Search
- Status transition
- Cancel
- Add note
- View payment
- View production
- View shipment

D. Workflow & State Behavior

Placed → payment/COD → confirmed → production → QC → packed → shipped → delivered
Exceptions: cancelled, failed delivery, return

E. Business Rules

- Only valid transitions
- Financial snapshot protected
- Cancellation policy state-dependent
- Customer cannot see internal notes

F. Exceptions / Error Handling

- Duplicate order
- Illegal transition
- Payment mismatch
- Cancellation not eligible

G. Security / Permissions

- Order ownership/role access
- Financial changes audited

H. Acceptance Criteria

- Unique order
- Complete timeline
- Invalid transitions blocked
- Customer history correct

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 10 — Custom Order Management

Special design/size/color orders with costing, quotation and customer approval.

A. Screens / Components

- Custom request
- Queue
- Detail
- Clarification
- Cost sheet
- Quote builder
- Quote versions
- Customer quote
- Accept/reject
- Convert to order

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Request ID	System	Yes	Unique	Module-specific validation; Unique
Product Type	Text/Ref	Yes	Required	Module-specific validation; Required
Size	Text	Yes	Unit-aware	Module-specific validation; Unit-aware
Color/Design	Text	Yes	Detailed	Module-specific validation; Detailed
Handle	Text	No	Optional	Module-specific validation; Optional
Quantity	Integer	Yes	>0	Module-specific validation; >0
Reference Image	File	No	Validated	Module-specific validation; Validated
Required Date	Date	No	Future	Module-specific validation; Future
Budget	Decimal	No	>=0	Module-specific validation; >=0
Material Cost	Decimal	Admin	Cost input	Module-specific validation; Cost input
Labour Cost	Decimal	Admin	Cost input	Module-specific validation; Cost input
Packaging Cost	Decimal	Admin	Cost input	Module-specific validation; Cost input
Shipping Cost	Decimal	Admin	Cost input	Module-specific validation; Cost input
Margin	Decimal	Admin	Business rule	Module-specific validation; Business rule
Quote Total	Decimal	System	Calculated	Module-specific validation; Calculated
Quote Version	Integer	System	Increment	Module-specific validation; Increment
Expiry	DateTime	Admin	Future	Module-specific validation; Future

C. User & System Actions

- Create
- Assign
- Clarify
- Cost
- Quote
- Revise
- Accept
- Reject
- Expire
- Convert

D. Workflow & State Behavior

Request → review → clarification → costing → quote → accept/reject → order → payment → production

E. Business Rules

- Accepted quote version frozen
- Revision requires reapproval
- Expired quote cannot be accepted
- All cost components auditable

F. Exceptions / Error Handling

- Incomplete spec
- Invalid image
- Expired quote
- Cost unavailable
- Post-acceptance change

G. Security / Permissions

- Customer sees own requests
- Costing restricted
- Quote actions audited

H. Acceptance Criteria

- Quote sent
- Customer accepts
- Order created
- Revision history preserved

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 11 — Wholesale / Bulk Orders

Bulk enquiries, quantity tiers, negotiated pricing and quotation conversion.

A. Screens / Components

- Wholesale landing
- Enquiry
- Queue
- Business profile
- Tier manager
- Quote
- Revision
- Acceptance
- Conversion

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Enquiry ID	System	Yes	Unique	Module-specific validation; Unique
Business Name	Text	Yes	Required	Module-specific validation; Required
Contact	Text	Yes	Required	Module-specific validation; Required
Mobile	String	Yes	Valid	Module-specific validation; Valid
Email	String	No	Valid	Module-specific validation; Valid
Product	Ref/Text	Yes	Required	Module-specific validation; Required
Quantity	Integer	Yes	>0	Module-specific validation; >0
Expected Price	Decimal	No	>=0	Module-specific validation; >=0
Delivery Location	Text	Yes	Required	Module-specific validation; Required
Required Date	Date	No	Valid	Module-specific validation; Valid
Tier	Reference	System	Pricing rule	Module-specific validation; Pricing rule
Unit Price	Decimal	System/Admin	>=0	Module-specific validation; >=0
Discount	Decimal	No	Validated	Module-specific validation; Validated
Shipping	Decimal	System/Admin	Rule	Module-specific validation; Rule
Total	Decimal	System	Calculated	Module-specific validation; Calculated
Validity	DateTime	Admin	Future	Module-specific validation; Future

C. User & System Actions

- Create enquiry
- Assign sales
- Select tier
- Override price
- Quote
- Revise
- Accept/reject
- Convert

D. Workflow & State Behavior

Enquiry → qualification → tier → costing → quote → acceptance → order → fulfillment

E. Business Rules

- Tier ranges cannot overlap
- Override permission required
- Quote revisions retained
- Expired quote renewal

F. Exceptions / Error Handling

- No tier
- Invalid quantity
- Expired quote
- Unauthorized override

G. Security / Permissions

- B2B data restricted
- Price override audited

H. Acceptance Criteria

- Tier correct
- Quote total correct
- Acceptance recorded
- Order retains wholesale price

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 12 — Inventory & Raw Materials

Finished goods, raw materials, packaging stock, reservation, consumption and ledger.

A. Screens / Components

- Finished stock
- Raw materials
- Inward
- Reserve
- Release
- Consume
- Adjustment
- Damage
- Return
- Ledger
- Low stock

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Inventory ID	System	Yes	Unique	Module-specific validation; Unique
Item Code	String	Yes	Unique	Module-specific validation; Unique
Item Name	Text	Yes	Required	Module-specific validation; Required
Type	Enum	Yes	Finished/Raw/Packaging	Module-specific validation; Finished/Raw/Packaging
Unit	Enum	Yes	kg/meter/piece	Module-specific validation; kg/meter/piece
Available	Decimal	System	>=0	Module-specific validation; >=0
Reserved	Decimal	System	>=0	Module-specific validation; >=0
Reorder Level	Decimal	No	>=0	Module-specific validation; >=0
Movement ID	System	Yes	Unique	Module-specific validation; Unique
Movement Type	Enum	Yes	Inward/Reserve/Consume/Adjust/Damage/Return	Module-specific validation; Inward/Reserve/Consume/Adjust/Damage/Return
Quantity	Decimal	Yes	>0	Module-specific validation; >0
Reference	String	Conditional	Traceable	Module-specific validation; Traceable
Reason	Text	Adjustment	Required	Module-specific validation; Required

C. User & System Actions

- Create item
- Receive
- Reserve
- Release
- Consume
- Adjust
- Damage
- Return
- View ledger
- Low-stock review

D. Workflow & State Behavior

Inward → available → reservation → consumption/release → adjustment/damage/return

E. Business Rules

- Negative stock blocked by default
- Adjustment requires reason
- Ledger corrections use reversal/adjustment
- Production consumption linked

F. Exceptions / Error Handling

- Insufficient stock
- Unknown item
- Invalid unit
- Missing reason
- Double consumption

G. Security / Permissions

- Role-restricted
- Ledger immutable after posting
- Adjustments audited

H. Acceptance Criteria

- Balances reconcile
- Consumption reduces stock
- Low-stock works
- Unauthorized adjustment blocked

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 13 — Production Management

Turns confirmed orders into assigned handmade production work.

A. Screens / Components

- Production queue
- Task detail
- Material plan
- Assignment
- Progress
- On hold
- Complete
- Rework

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Production ID	System	Yes	Unique	Module-specific validation; Unique
Order ID	Reference	Yes	Confirmed order	Module-specific validation; Confirmed order
Product/Variant	Reference	Yes	Snapshot	Module-specific validation; Snapshot
Planned Qty	Integer	Yes	>0	Module-specific validation; >0
Assigned Staff	Reference	No	Production role	Module-specific validation; Production role
Required Date	Date	No	Valid	Module-specific validation; Valid
Material Plan	List	No	Expected	Module-specific validation; Expected
Issued Material	List	No	Actual	Module-specific validation; Actual
Completed Qty	Integer	System	Controlled	Module-specific validation; Controlled
Status	Enum	Yes	Pending/Assigned/In Production/Completed/On Hold/Rework	Module-specific validation; Pending/Assigned/In Production/Completed/On Hold/Rework
On Hold Reason	Text	Conditional	Required	Module-specific validation; Required

C. User & System Actions

- Create
- Assign
- Issue material
- Start
- Pause
- Progress
- Complete
- Rework

D. Workflow & State Behavior

Confirmed order → task → assignment → material issue → production → complete → QC
QC rework → production

E. Business Rules

- Only confirmed orders
- Material issue traceable
- Over-completion controlled
- On-hold reason required

F. Exceptions / Error Handling

- Insufficient material
- No staff
- Overproduction
- Cancelled order

G. Security / Permissions

- Production access role
- Material usage audited

H. Acceptance Criteria

- Task linked
- Progress visible
- Completion enters QC
- Rework traceable

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 14 — Quality Control & Packing

Inspects handmade quality and prepares packages for shipment.

A. Screens / Components

- QC queue
- Inspection
- Checklist
- QC history
- Rework
- Packing queue
- Package builder
- Packed confirmation

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
QC ID	System	Yes	Unique	Module-specific validation; Unique
Order/Production	Reference	Yes	Linked	Module-specific validation; Linked
Size	Enum	Yes	Pass/Fail	Module-specific validation; Pass/Fail
Color/Design	Enum	Yes	Pass/Fail	Module-specific validation; Pass/Fail
Weaving/Finish	Enum	Yes	Pass/Fail	Module-specific validation; Pass/Fail
Handle	Enum	No	Pass/Fail	Module-specific validation; Pass/Fail
Quantity	Integer	Yes	Count	Module-specific validation; Count
Result	Enum	Yes	PASS/FAIL/REWORK	Module-specific validation; PASS/FAIL/REWORK
Remarks	Text	Conditional	Required on fail	Module-specific validation; Required on fail
Package Count	Integer	Packing	>0	Module-specific validation; >0
Package Weight	Decimal	Packing	If required	Module-specific validation; If required
Dimensions	Text	No	Shipping data	Module-specific validation; Shipping data

C. User & System Actions

- Inspect
- Pass
- Fail
- Rework
- Reinspect
- Create package
- Edit package
- Mark packed

D. Workflow & State Behavior

Production complete → QC → pass → packing → packed
Fail/rework → production → QC

E. Business Rules

- Shipment blocked unless QC pass
- Packing blocked before pass
- Finalized QC changes audited
- Package data required for shipping

F. Exceptions / Error Handling

- Missing checklist
- Quality fail
- Quantity mismatch
- Packing before QC

G. Security / Permissions

- QC/packing roles restricted
- Inspection history protected

H. Acceptance Criteria

- Pass permits packing
- Fail blocks shipping
- Rework linked
- Package details stored

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 15 — Shipping & Delivery

Shipment creation, courier/tracking, delivery states, failure and return.

A. Screens / Components

- Ready queue
- Shipment
- Courier
- Tracking
- Timeline
- Delivery update
- Failed delivery
- Return

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Shipment ID	System	Yes	Unique	Module-specific validation; Unique
Order ID	Reference	Yes	Linked	Module-specific validation; Linked
Courier	Text	Conditional	Required when shipped	Module-specific validation; Required when shipped
Tracking Number	String	Conditional	Carrier rule	Module-specific validation; Carrier rule
Ship Date	DateTime	System	Timestamp	Module-specific validation; Timestamp
Expected Delivery	Date	No	Optional	Module-specific validation; Optional
Status	Enum	Yes	Ready/Shipped/In Transit/OFD/Delivered/Failed/Returned	Module-specific validation; Ready/Shipped/In Transit/OFD/Delivered/Failed/Returned
Failure Reason	Text	Conditional	Required on failure	Module-specific validation; Required on failure
Package Count	Integer	Yes	From packing	Module-specific validation; From packing
Weight	Decimal	No	From packing	Module-specific validation; From packing

C. User & System Actions

- Create shipment
- Assign courier
- Add tracking
- Ship
- Transit
- OFD
- Deliver
- Fail
- Return

D. Workflow & State Behavior

Packed → Ready → Shipped → In Transit → OFD → Delivered
Failure → retry/return

E. Business Rules

- Only packed orders ship
- Tracking uniqueness
- Delivered reversal controlled
- Failure requires reason

F. Exceptions / Error Handling

- Missing tracking
- Courier unavailable
- Delivery failure
- Return
- Duplicate shipment

G. Security / Permissions

- Shipping role restricted
- Status audited

H. Acceptance Criteria

- Tracking visible
- Order sync
- Failure creates exception

- Return distinct

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 16 — Customer Engagement

Reviews, notifications, WhatsApp, offers and coupons.

A. Screens / Components

- Review form
- Moderation
- Notification center
- WhatsApp product CTA
- Custom WhatsApp
- Wholesale WhatsApp
- Coupon manager
- Offer manager

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Review ID	System	Yes	Unique	Module-specific validation; Unique
Order/Product	Reference	Yes	Eligibility	Module-specific validation; Eligibility
Rating	Integer	Yes	Configured scale	Module-specific validation; Configured scale
Review Text	Text	No	Length limit	Module-specific validation; Length limit
Images	Files	No	Validated	Module-specific validation; Validated
Coupon Code	String	Yes	Unique	Module-specific validation; Unique
Discount Type	Enum	Yes	Flat/Percentage	Module-specific validation; Flat/Percentage
Discount Value	Decimal	Yes	≥ 0	Module-specific validation; ≥ 0
Minimum Order	Decimal	No	≥ 0	Module-specific validation; ≥ 0
Validity	DateTime	Yes	Valid range	Module-specific validation; Valid range
Usage Limit	Integer	No	≥ 0	Module-specific validation; ≥ 0

C. User & System Actions

- Submit
- Moderate
- Publish/hide
- Create/deactivate coupon
- Validate coupon
- Send notification
- Open WhatsApp

D. Workflow & State Behavior

Delivered → review eligibility → submit → moderation → publish
Coupon → validate → discount

E. Business Rules

- Review eligibility enforced
- Coupon server-side
- WhatsApp uses configured business number
- Notification failures tracked

F. Exceptions / Error Handling

- Expired coupon
- Usage exceeded
- Ineligible review
- WhatsApp misconfiguration
- Notification failure

G. Security / Permissions

- Review ownership protected
- Promotion changes audited

H. Acceptance Criteria

- Review works
- Coupon cannot bypass totals
- WhatsApp context correct
- Notifications generated

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 17 — Admin Dashboard & Reports

Management KPIs, operational queues, filters, drill-down and exports.

A. Screens / Components

- Dashboard
- Sales
- Orders
- Payments
- UPI queue
- COD
- Inventory
- Production
- QC
- Packing
- Shipping
- Customers
- Custom
- Wholesale
- Exports

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Date From	Date	Filter	Valid	Module-specific validation; Valid
Date To	Date	Filter	>= from	Module-specific validation; >= from
Order Status	Enum	Filter	Configured	Module-specific validation; Configured
Payment Method	Enum	Filter	UPI/COD	Module-specific validation; UPI/COD
Product/Category	Reference	Filter	Optional	Module-specific validation; Optional
Customer	Reference	Filter	Permission-aware	Module-specific validation; Permission-aware
Report Type	Enum	Yes	Configured	Module-specific validation; Configured
Export	Enum	No	CSV/XLSX/PDF if implemented	Module-specific validation; CSV/XLSX/PDF if implemented

C. User & System Actions

- Filter
- Generate
- Drill-down
- Open source
- Export

D. Workflow & State Behavior

Dashboard → report → filters → aggregate → drill-down → export

E. Business Rules

- Reports use authoritative records
- Cancelled/returned treatment defined
- Permissions apply
- Exports respect filters

F. Exceptions / Error Handling

- No data
- Large query
- Unauthorized report
- Export error

G. Security / Permissions

- Sensitive reports restricted
- Export logging where required

H. Acceptance Criteria

- Dashboard reconciles
- Filters work
- Drill-down accurate
- Export matches report

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

Module 18 — System Settings, Security & Audit

Business configuration, payment settings, roles, permissions, security and traceability.

A. Screens / Components

- Business profile
- UPI settings
- COD
- Shipping
- Tax/GST if applicable
- Order numbering
- Roles
- Permissions
- Notifications
- Audit log

B. Field-Level Functional Specification

Field	Type	Required	Source / Default	Validation / Behavior
Business Name	Text	Yes	Required	Module-specific validation; Required
Logo	File	No	Validated	Module-specific validation; Validated
UPI ID	String	Conditional	If UPI enabled	Module-specific validation; If UPI enabled
UPI QR	File	Conditional	If UPI enabled	Module-specific validation; If UPI enabled
UPI Enabled	Boolean	Yes	Toggle	Module-specific validation; Toggle
COD Enabled	Boolean	Yes	Toggle	Module-specific validation; Toggle
Shipping Rules	Structured	Yes	Validated	Module-specific validation; Validated
Tax/GST	Structured	No	Business-approved	Module-specific validation; Business-approved
Order Prefix	Text	No	Configured	Module-specific validation; Configured
Role	Text	Yes	Unique	Module-specific validation; Unique
Permission	Enum	Yes	Configured	Module-specific validation; Configured
Audit Event	System	Yes	Immutable log	Module-specific validation; Immutable log

C. User & System Actions

- Update business profile
- Enable/disable UPI
- Replace QR
- Enable/disable COD
- Shipping rules
- Manage roles
- Review audit
- Notification settings

D. Workflow & State Behavior

Edit → validate → authorize → save → audit → effective
Historical orders/payments retain historical values

E. Business Rules

- Disabling payment method affects new orders only
- Audit logs protected
- UPI change strongly authorized
- Secrets not in code

F. Exceptions / Error Handling

- Invalid UPI
- Missing QR
- Invalid shipping rule
- Unauthorized settings
- Role lockout

G. Security / Permissions

- Settings APIs protected
- Critical changes audited
- Payment proof/customer data access controlled

H. Acceptance Criteria

- Checkout reflects settings
- Historical records preserved
- Audit complete
- No deployment needed for routine settings

I. QA Test Focus

- Positive, negative and boundary-value testing.
- Role/permission testing.
- Duplicate/retry and refresh behavior.
- Status transition and persistence testing.
- Responsive behavior for customer-facing screens.
- Audit/log verification for sensitive actions.

19. Complete Standard Retail Customer Journey

Customer lands on Home → category/search → product detail → selects variant → cart → checkout → address → shipping calculation → UPI or COD → final validation → order created. UPI: external payment → screenshot/UTR → verification → approve/reject. Approved UPI/COD → production → QC → packing → shipping → delivery → review/support.

Journey Acceptance

- Only UPI/COD are displayed.
- Final amount equals server calculation.
- Order is created exactly once.
- UPI proof never equals automatic payment.
- Order can be traced through every operational stage.

20. Complete UPI Manual Payment Journey

Step	Customer	System	Admin
1	Select UPI	Show QR, UPI ID, amount	—
2	Pay externally	Keep pending	—
3	Upload proof	Validate/store	—
4	Submit	Under Review	Queue
5	—	Verification pending	Open proof/amount/UTR
6	—	Await decision	Approve or Reject
7A	—	Paid + confirmed	Approve with actor/time
7B	Resubmit	Rejected/Resubmission	Reason mandatory

UPI Safety Controls

- Screenshot is evidence, not automatic confirmation.
- Admin compares against actual payment records.
- Duplicate UTR is flagged where UTR is used.
- Approved payment is protected from ordinary edits.
- Every decision is audited.

21. Complete COD Journey

- COD eligibility checked.
- Customer selects COD.
- COD payment state created.
- No screenshot requested.
- Order confirmation follows policy.
- Order enters production/fulfillment.
- Delivery/cash reconciliation is operationally reportable.

22. Complete Custom Order Journey

Stage	Customer	Admin/Sales	System
Request	Specs + image	Review	Request ID
Clarification	Answer	Ask	Conversation/status log
Costing	—	Material + labour + packaging + shipping + margin	Calculate
Quote	View	Send/revise	Version
Decision	Accept/reject	View	Freeze accepted version
Order	Pay UPI/COD	Confirm	Order snapshot
Fulfillment	Track	Plan/assign	Production → QC → Packing → Shipping

23. Complete Wholesale Journey

- Capture business identity, product, quantity, delivery location and required date.
- Select applicable quantity tier.
- Allow authorized negotiated price override.
- Issue quote with unit price, discount, shipping, total and validity.
- Customer accepts.

- Accepted quote converts to order with price snapshot.
- Fulfill from stock or production.
- Maintain traceability from enquiry to quote to order.

24. Order State Machine

Current	Allowed Next	Gate
ORDER_PLACED	PAYMENT_PENDING / COD_CONFIRMED / CANCELLED	Payment method
PAYMENT_PENDING	PROOF_SUBMITTED / CANCELLED	Customer
PROOF_SUBMITTED	UNDER_REVIEW	Proof
UNDER_REVIEW	PAYMENT_CONFIRMED / PAYMENT_REJECTED	Admin
PAYMENT_REJECTED	PROOF_SUBMITTED / CANCELLED	Resubmit
COD_CONFIRMED	ORDER_CONFIRMED / CANCELLED	COD policy
PAYMENT_CONFIRMED	ORDER_CONFIRMED	Approved payment
ORDER_CONFIRMED	PRODUCTION / CANCELLED	Operations
PRODUCTION	QC	Complete
QC	PACKED / REWORK	Result
REWORK	PRODUCTION	Rework
PACKED	SHIPPED	Shipment
SHIPPED	IN_TRANSIT / DELIVERY_FAILED	Carrier
IN_TRANSIT	OUT_FOR_DELIVERY / DELIVERY_FAILED	Carrier
OUT_FOR_DELIVERY	DELIVERED / DELIVERY_FAILED	Carrier
DELIVERY_FAILED	SHIPPED / RETURNED	Resolution
DELIVERED	RETURN_REQUESTED	Policy
RETURN_REQUESTED	RETURNED / REJECTED	Decision

25. Data Entity Catalogue

Entity	Purpose	Relationships
User	Identity	Role, customer, audit
Role/Permission	Authorization	User → Role → Permissions
Address	Delivery	Customer → addresses; order snapshot
Category	Catalogue	Category → products
Product	Sellable master	Category, variants, media, inventory
Variant	Options	Product → variants
Cart/Cart Item	Temporary purchase	Customer/session → items
Order/Order Item	Transaction	Customer, snapshots, payment, shipment
Payment/Proof	Payment state/evidence	Order → payment attempts
Custom Request/Quote	Special orders	Customer → quote → order
Wholesale/Quote	Bulk sales	Business → quote → order
Inventory/Movement	Stock	Product/material → ledger
Production	Work	Order → production
QC	Inspection	Production/order → QC
Packing	Packages	Order → packing
Shipment	Delivery	Order → shipment
Review	Feedback	Customer → product/order
Coupon/Offer	Promotion	Rule → cart/order
Notification	Communication	User/event
Audit Log	Traceability	Actor/entity/action

26. Business Rule Catalogue

- BR-001 Only manual UPI verification and COD are supported initially.
- BR-002 Screenshot upload never automatically marks payment as paid.
- BR-003 UPI is paid only after authorized approval.
- BR-004 Rejection requires reason.
- BR-005 Rejected proof may be resubmitted.

- BR-006 COD requires no payment proof.
- BR-007 Final totals are server-calculated.
- BR-008 Price and stock are revalidated before order creation.
- BR-009 Confirmed order financial snapshots remain stable.
- BR-010 Invalid order transitions are blocked.
- BR-011 QC failure blocks shipping.
- BR-012 Packing requires QC pass.
- BR-013 Shipping normally requires packed state.
- BR-014 Custom special pricing uses quotation.
- BR-015 Wholesale quantity pricing is configurable.
- BR-016 Inventory adjustments require permission and reason.
- BR-017 Critical admin actions are audited.
- BR-018 Payment method availability controlled by settings.
- BR-019 Disabling payment method preserves history.
- BR-020 Customer cannot access admin functions.
- BR-021 Quote acceptance freezes quote version.
- BR-022 Expired quote requires renewal.
- BR-023 Production completion feeds QC.
- BR-024 QC rework returns to production.
- BR-025 Catalogue changes do not rewrite historical orders.

27. Validation & Error Matrix

Area	Validation	Expected Response
Authentication	Invalid credentials	Reject safely; rate-limit repeated attempts
Catalog	Duplicate SKU	Block save
Catalog	Invalid price	Block save
Variant	Unavailable combination	Block selection
Cart	Quantity > stock	Require correction
Checkout	Missing address	Block order
Checkout	Disabled method	Hide/reject
UPI	Missing proof	Cannot submit
UPI	Reject without reason	Block decision
UPI	Wrong amount	Flag/reject per policy
Order	Illegal transition	Reject/log
Custom	Expired quote	Block acceptance
Wholesale	No tier	Route to quotation
Inventory	Insufficient stock	Block movement
Production	Over-completion	Block/authorize
QC	Missing checklist	Cannot finalize
Packing	No QC pass	Cannot pack
Shipping	Not packed	Cannot ship
Coupon	Expired	Reject
Reports	Unauthorized	Deny

28. Notification Matrix

Event	Customer	Admin/Sales	Production/QC
Order placed	Yes	Yes	No
UPI proof submitted	Status	Yes	No
UPI approved	Yes	Yes	No
UPI rejected	Yes + reason	Yes	No
Order confirmed	Yes	Yes	Configured
Custom quote	Yes	Yes	No
Wholesale quote	Yes	Yes	No
Production assigned	No	Yes	Yes
QC required	No	Yes	Yes
Packed	Yes	Yes	No

Event	Customer	Admin/Sales	Production/QC
Shipped	Yes + tracking	Yes	No
Delivered	Yes	Yes	No
Delivery failed	Yes	Yes	No

29. Reports & KPI Definitions

Report	Filters	Output
Sales	Date/product/category/status	Orders, qty, values, discounts, shipping
Orders	Date/status/customer	Count, value, ageing
UPI	Date/status	Pending, approved, rejected, ageing
COD	Date/status	Orders, delivery, reconciliation
Inventory	Item/type/date	Opening/inward/reserve/consume/closing
Low Stock	Item/category	Below reorder level
Production	Date/status/staff	Pending/active/completed/ageing
QC	Date/result	Pass/fail/rework
Shipping	Date/status/courier	Lifecycle
Custom	Date/status	Request/quote/order funnel
Wholesale	Date/status	Enquiry/quote/order funnel

30. File Upload & Media Requirements

File	Use	Controls
Product image	Catalogue	Type/size validation, optimization
Custom reference	Design	Restricted access, validation
UPI screenshot	Payment	Restricted access, secure storage
Review image	Review	Validation/moderation
UPI QR	Payment setting	Admin replacement, checkout-readable

- Extension alone is not a security check.
- Safe storage names.
- No executable files.
- Sensitive proof URLs not publicly guessable.
- Historical files retained according to policy.

31. Security Requirements

- HTTPS.
- Secure password hashing.
- Server-side authorization.
- Rate limiting.
- Input validation/output encoding.
- Secure session/token handling.
- Secrets outside source code.
- Payment proof access control.
- Audit critical actions.
- Backup/restore testing.
- Least privilege.

32. Non-Functional Requirements

Area	Requirement
Performance	Define response budgets during technical design; optimize storefront/admin.
Availability	Monitoring, backup and recovery.
Scalability	Support increasing products, orders, customers and staff.
Responsive UX	Customer journey usable on desktop/tablet/mobile.
Accessibility	Labels, focus, contrast and validation.
Observability	Critical errors/workflow failures monitored.
Privacy	Customer/payment proof access controlled.

Area	Requirement
Auditability	Critical actions traceable.
Maintainability	Business rules documented/configurable.

33. QA / UAT Scenario Catalogue

- Authentication: registration, duplicates, invalid password, reset, suspension, role access.
- Catalogue: category, SKU, price, MOQ, media, activation, historical order.
- Variants: valid/invalid combinations, custom size/color, image upload.
- Cart: quantity, MOQ, stock change, price change, duplicate requests.
- Checkout: address, shipping, coupon, disabled UPI/COD, final total.
- UPI: QR, ID, proof, UTR, submit, review, approve, reject, resubmit, wrong amount, duplicate UTR.
- COD: enabled/disabled, order creation, no proof.
- Orders: lifecycle, cancellation, duplicate, invalid transitions.
- Custom: request, clarification, costing, quote, expiry, revision, acceptance.
- Wholesale: enquiry, tier, override, quote, acceptance.
- Inventory: inward, reserve, release, consume, adjustment, insufficient stock.
- Production: assignment, material issue, progress, completion, rework.
- QC/Packing: pass, fail, rework, package details, shipment gating.
- Shipping: shipment, tracking, transit, delivery, failure, return.
- Engagement: review, moderation, coupon, notifications, WhatsApp.
- Security: authorization bypass, upload abuse, rate limit, sensitive-data exposure, audit integrity.

34. Definition of Done

- Business owner approval.
- UI/UX coverage for every required screen.
- Server-side validation and authorization.
- UPI fully tested.
- COD fully tested.
- State machine tested.
- Custom/wholesale quotation tested.
- Inventory → production → QC → packing → shipping tested.
- Reports reconciled.
- Audit verified.
- Backup/restore tested.
- Security testing completed.
- UAT signed off.

35. Open Business Decisions Before Development

Decision	Impact
UPI UTR mandatory/optional	Proof validation
UPI verification SLA	Customer messaging/ageing
COD eligibility	Checkout rules
Shipping charge model	Final total
Cancellation window	Order states
Return/replacement policy	Post-delivery workflow
Tax/GST	Pricing/invoice
Invoice format	Documents
Quote validity	Acceptance
Wholesale tiers	Pricing
Production costing formula	Custom quotes
Courier process	Manual vs integration
Notification channels	Email/SMS/WhatsApp/web
Data retention	Proof/audit/customer records

36. Final Functional Principle

Deepa Handmade operates as one connected system: storefront → cart → checkout → UPI/COD → order → inventory/production → QC → packing → shipping → delivery → engagement → reporting. Standard products remain simple for customers; custom and wholesale business uses controlled quotation workflows; operational and payment actions remain traceable.

END OF EXPANDED DETAILED FRD v3.0

Deepa Handmade — Functional Requirements Reference